

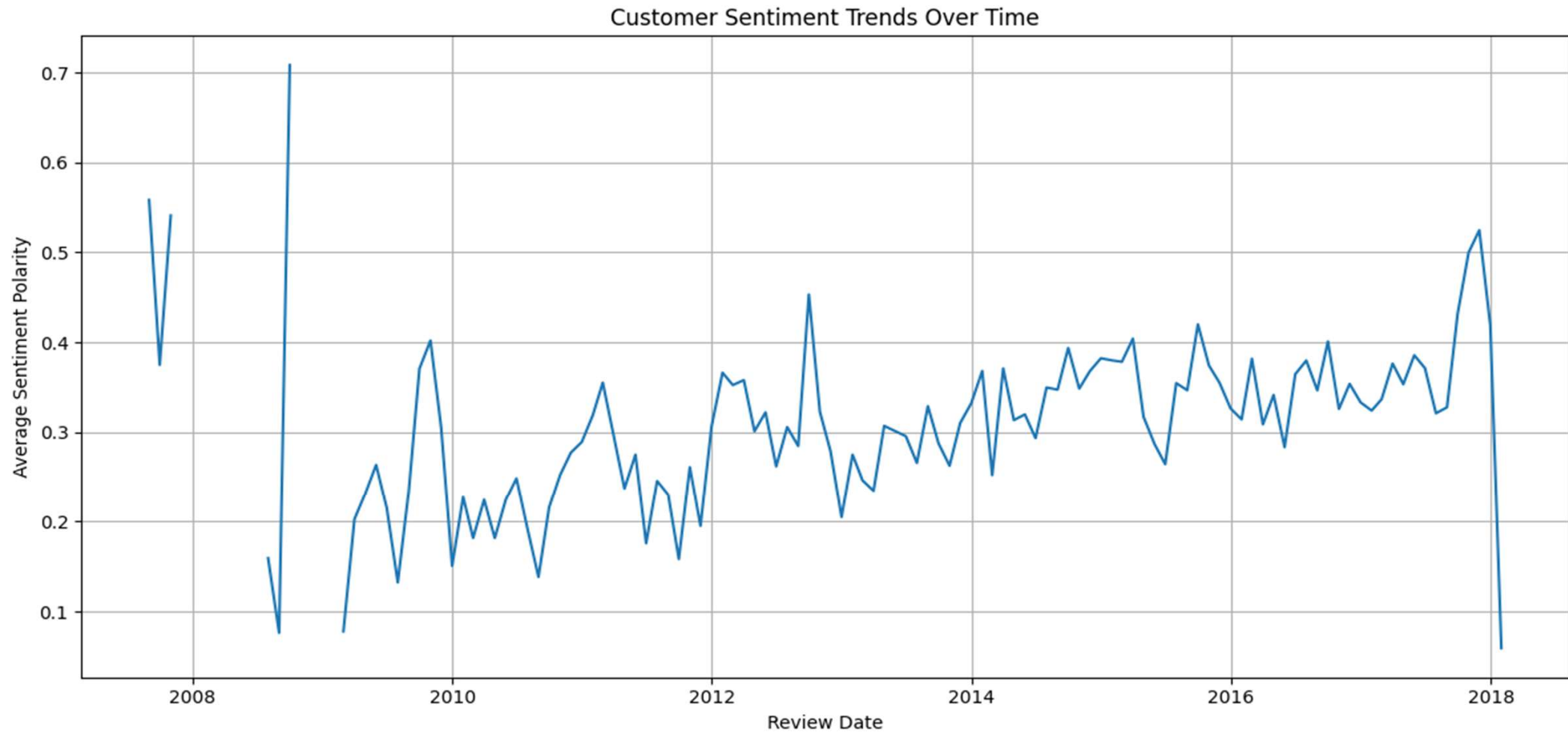
Tyree N. Seals

Capstone 3

LSU Fullstack

A/I & M/L

cohort- 2601-ftb-ct-aim-pt-b



2007-2009 had more variability because of fewer reviews.
2012-2017 customer reviews were stable, and the customers seem to be happy.
The dramatic drop in 2018 is from customers becoming unhappy.

Logistic Regression	Precision	Recall	F1-Score	Support
0.0	0.50	0.89	0.64	227
1.0	0.98	0.89	0.93	1774
Accuracy			0.89	2001
Macro avg	0.74	0.89	0.79	2001
Weighted avg	0.93	0.89	0.90	2001
0.8866				

CLASS 1(FAKE REVIEWS) : 98% was a fake review. The model successfully found 89% of the model were truly fake.

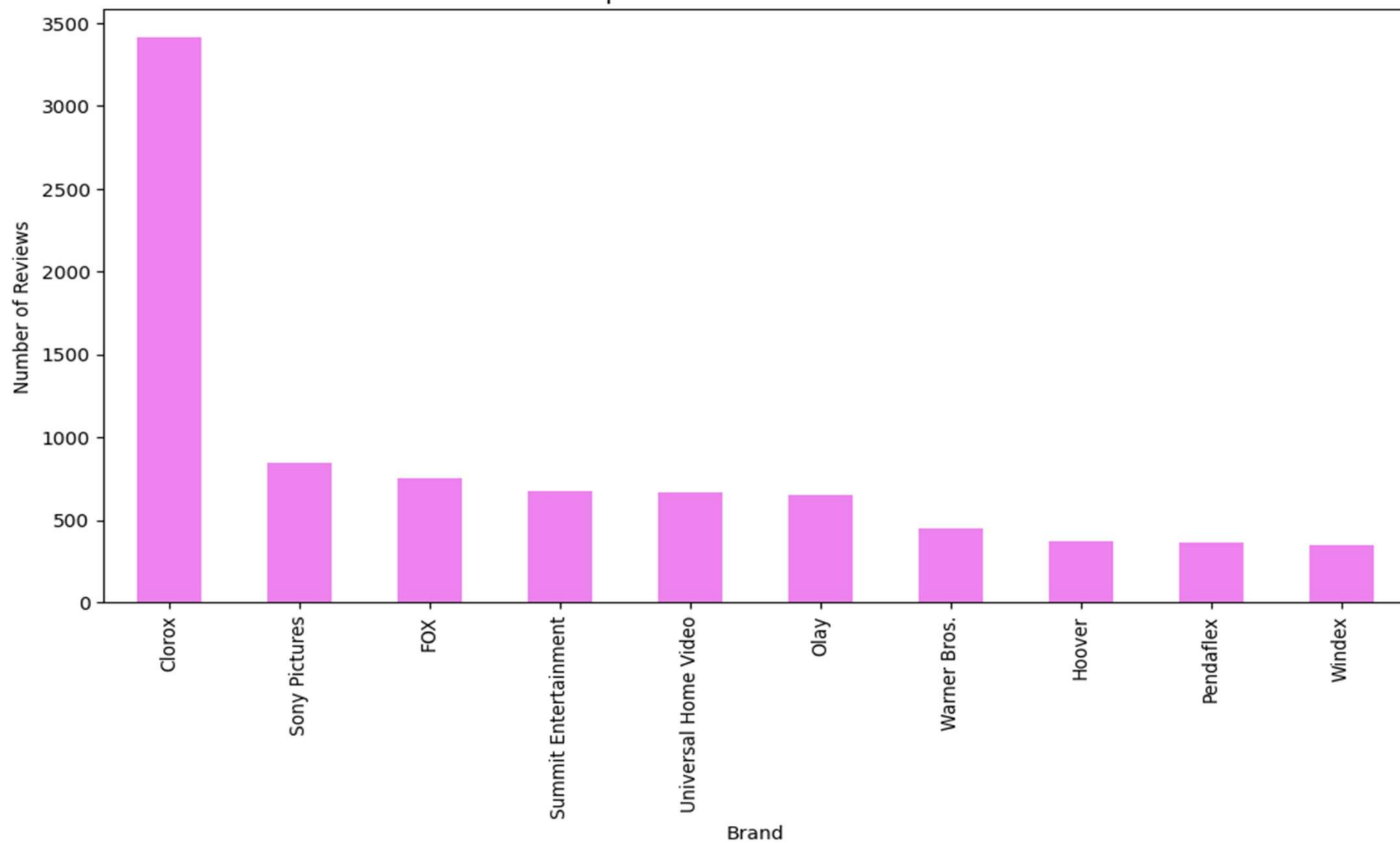
CLASS 2(REAL REVIEWS): only 50% were predicted as real, the model is 89% sure this is a fact.

64% F1 score is weak, but I made a proxy label because of the imbalanced data set, so I reran this model a second time.

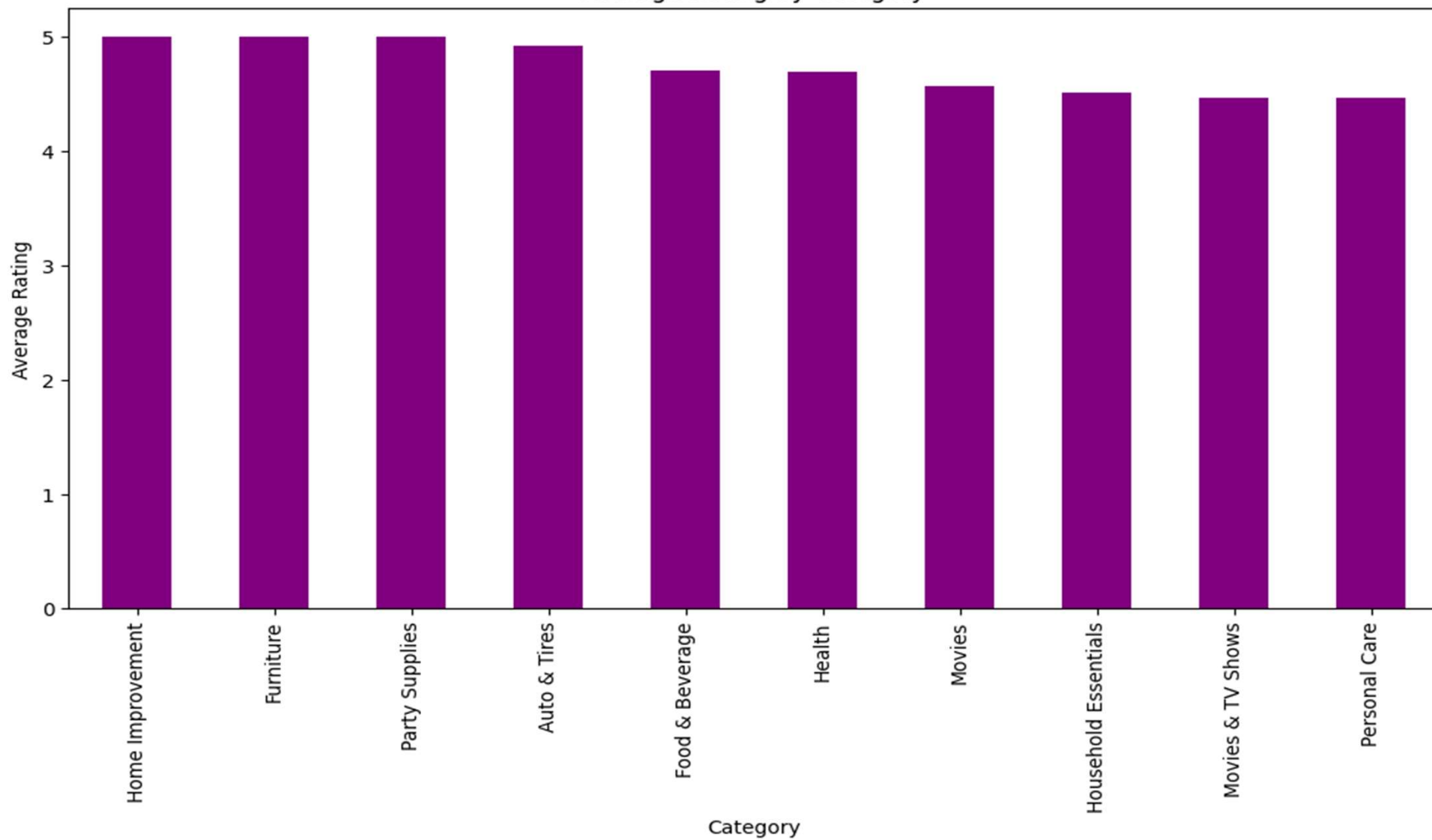
ISOLATION FOREST

- Found 10% as anomalies and 89.99% as normal reviews.
- FOR REAL REVIEWS: 446 in total
- FOR FAKE REVIEWS 3,798 in Total

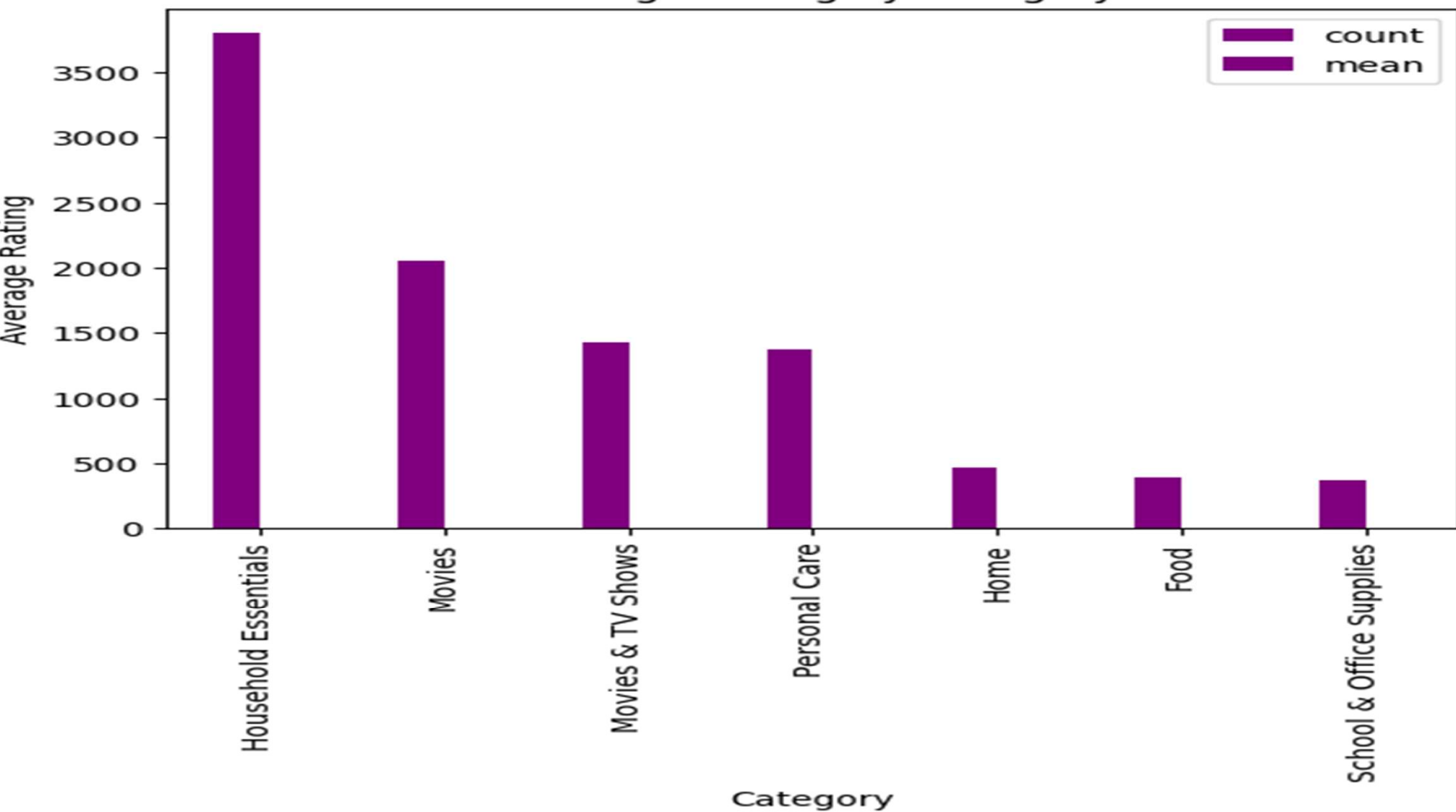
Top 10 Most Reviewed Brands



Average Rating by Category



Average Rating by Category



Decision Tree	Precision	Recall	F1 Score	Support
0	0.21	0.81	0.33	89
1	0.97	0.63	0.76	760
Accuracy			0.65	849
Macro avg	0.59	0.72	0.55	849
Weighted avg	0.89	0.65	0.72	849
Real Review	0.21	0.81	0.33	89
Fake Review	0.97	0.63	0.76	760
Accuracy			0.65	849
Macro avg	0.59	0.72	0.55	849
Weighted avg	0.89	0.65	0.72	849
Confusion Matrix:				
[[72 17] [279 481]]				

Decision tree was correct 65% of the time and has a low F1 score 33% of real reviews. This is a poor model, and its probably the data set isn't balance, and it tried to over fit in the training. Decision tree performs with low effectiveness, it works but not so great.

RANDOM FOREST MODEL

	Precision	Recall	F1-Score	Support
Real Review	0.50	0.46	0.48	89
Fake Review	0.94	0.95	0.94	760
Accuracy			0.90	849
Macro avg	0.72	0.70	0.71	849
Weighted avg	0.89	0.90	0.89	849
MODEL:	Accuracy	Precision	Recall	F1 Score
0 Random Forest	0.8952	0.9374	0.9461	0.9417

RANDOM Forest did GREAT! It was correct 89.52 % of the time, and as a F1 score of 94.17, so this Model is excellent at figuring out which reviews are fake.

Logistic REGRESSION MODEL

Model	Accuracy	Precision	Recall	F1 Score
0 Logistic Regression	0.8504	0.9648	0.8645	0.9119

This model is another High performer, with an F1 score of 91.19% it's great to use for finding fake reviews in the dataset

Model: RNN	Precision	Recall	F1 Score	Support
Real Review	0.37	0.69	0.48	89
Possible Fake Review	0.96	0.86	0.91	760
Accuracy			0.84	849
Macro avg	0.66	0.77	0.69	849
Weighted avg	0.90	0.84	0.86	849
0.8433451118963486				

RNN got an accuracy of 84.33%, the 48% F1 score on the real reviews is due to the data being skewed, I made it biased towards fake reviews, so it's imbalanced. It's not the best models, but it's ok, it did better than Decision tree.

DistilBERT Model	Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
	1	0.231800	0.213378	0.906667	0.868881	0.906667	0.876391
	2	0.205600	0.192611	0.928887	0.915235	0.926667	0.914750
	3	0.150200	0.164950	0.946667	0.942191	0.946667	0.942414

DISTIL BERT got better with every epoch, and did great at finding fake reviews. The F1 was over 94%. It takes a long time to run this model, but it's worth it in the end. This model gets better when you give it more, and have a better computer.

Concatenated Data- Logistic Regression, Decision Tree, Random Forest, BERT.

Which model found the fake reviews the best?

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.8866	0.9800	0.8900	0.9380
1	Decision Tree	0.6514	0.9659	0.6329	0.7647
2	Random Forest	0.8952	0.9374	0.9461	0.9417
3	BERT	0.9467	0.9422	0.9467	0.9424



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It's BERT !!!!! From Sesame Street

The F1 scores don't lie, BERT came in 1st and Random Forest 2nd with finding those fake reviews.

A CLOSER LOOK AT COCATENATE

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I WILL upload in the chatbox an app created with GRADIO that shows these findings.

- Use this data to summarize reviews for multiple companies.
- I used BERT to Gradio for this one, Thank you everyone!

References:

- Professor Cristopher
 - GOOGLE
 - GEMINI
- FULLSTACK assignment models!
 - Bert from Sesame Street